Understanding Trajectron++: A Per-Trajectory Performance Analysis

Simon Drauz and Zoe Matrullo

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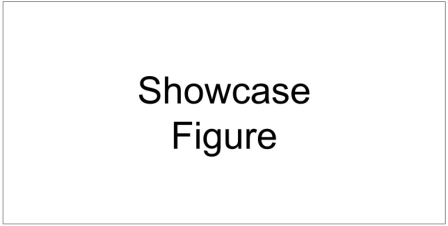

Outline

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Trajectory Prediction for Autonomous Driving

- **Task:** forecast future (x, y) positions of agents (pedestrians, cyclists, vehicles) from observed past movements

[figure placeholder]

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- Challenging due to **complex agent interactions** and influence of **static context**
- Essential for **safe path planning** and **collision avoidance** in autonomous vehicles

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  - Outputs a distribution over future trajectories
- Handles *heterogeneous* agent types jointly (pedestrian, bicycle, car) and integrates HD map context

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Prediction metrics:

`ml_ade`	most-likely ADE ← **used**
`ml_fde`	most-likely FDE
`min_ade_5`	best-of-5 ADE
`nll_mean`	negative log-likelihood

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2. What are the *success and failure modes* that explain model performance?

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1. Which trajectory and scene characteristics affect model performance?
2. What are the *success and failure modes* that explain model performance?
3. How do model settings (history window, prediction horizon, attention radius) affect performance?

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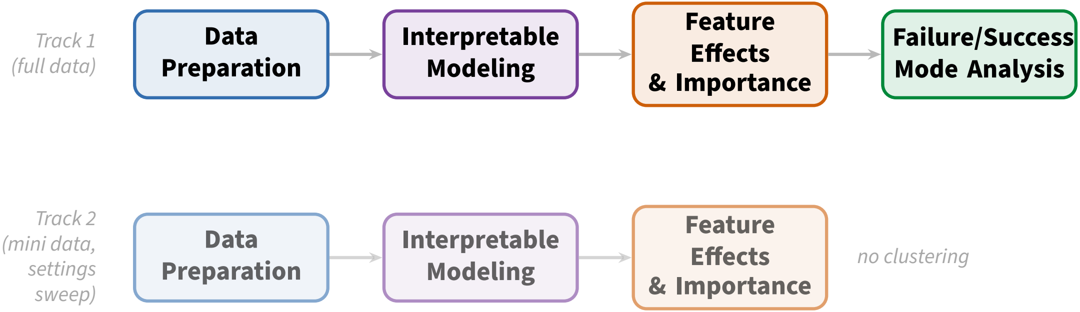

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Standard Trajectron++

- Evaluation outputs *batch-averaged* ADE / FDE

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- `data_idx`: stable trajdata identifier, used to join evaluation output with computed trajectory & scene characteristics

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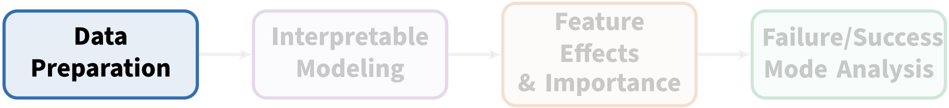

Feature Engineering — Trajectory & Scene Characteristics

Trajectron++ outputs only (x, y) position sequences — all features below are computed by us from those trajectories.

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$\Rightarrow$ **18 candidate features** joined with Trajectron++ evaluation output via `data_idx`

Target Distribution & Log Transformation

- `ml_ade` is strongly *right-skewed* (skewness $= 2.59$)

[figure placeholder: before/after distribution]

Target Distribution & Log Transformation

- `ml_ade` is strongly *right-skewed* (skewness $= 2.59$)
- Log transformation (`log1p`) considered to stabilise variance — tested as a modeling choice in the next step

[figure placeholder: before/after distribution]

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`std_speed`, `mean_acceleration`, `min_neighbor_distance`, `heading_change_per_sec`, `scene_num_VEHICLE`, `scene_density_VEHICLE`

Retains scene-level features alongside motion features.

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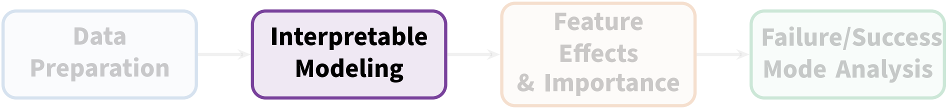

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  - **GAM:** smooth effects directly readable from the model
  - **XGBoost:** post-hoc via SHAP values and partial dependence plots
- This turns ADE into **actionable understanding** of model behaviour

GAM and XGBoost

GAM (Generalized Additive Model)

Inherently interpretable

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Black-box, interpreted post-hoc

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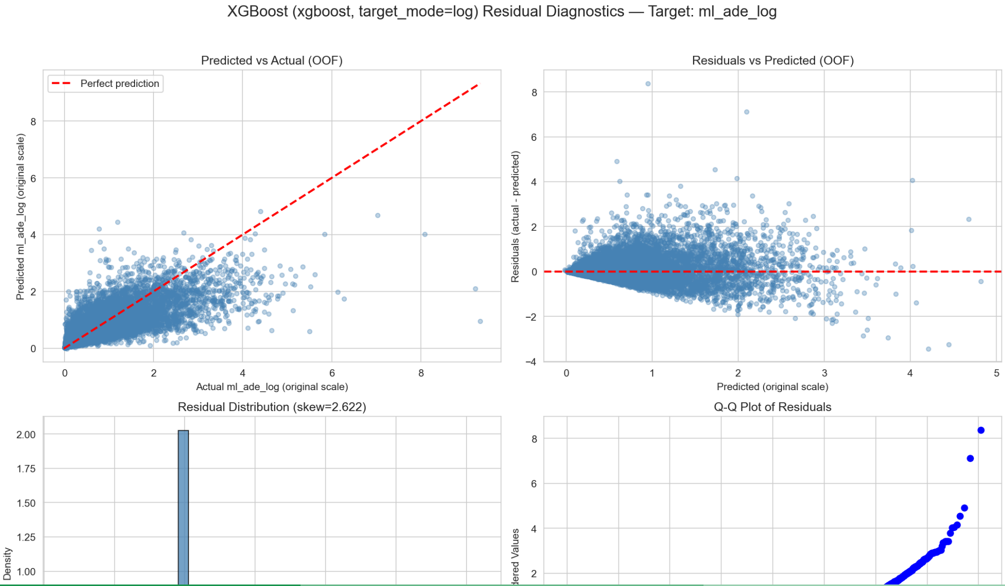
- Interpreted via **SHAP values** and **partial dependence plots**

Validating the Tuning Procedure — Nested Resampling

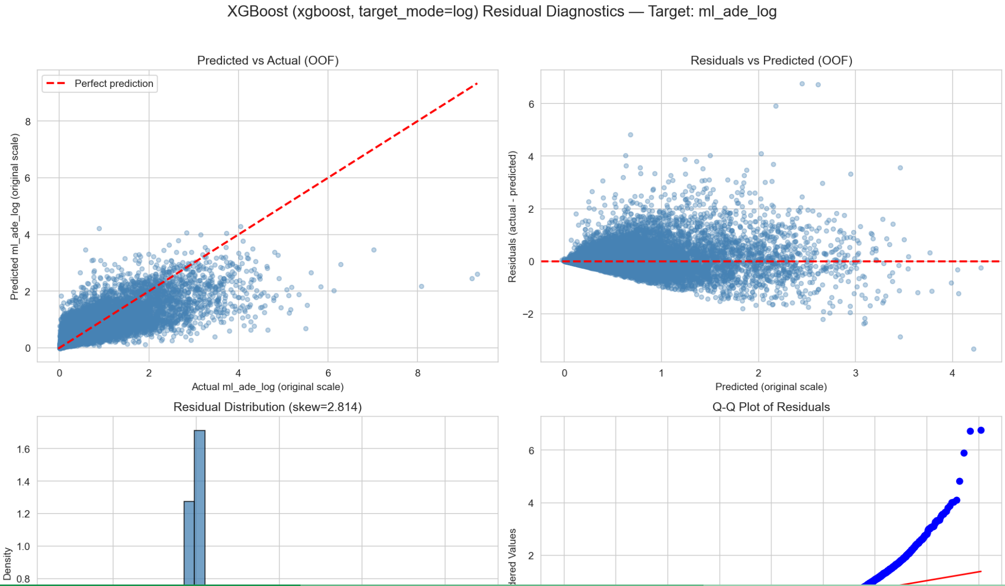
- **Problem:** hyperparameter tuning risks overfitting to a particular data split
- **Solution:** nested cross-validation
  - Outer folds: evaluate generalization of the full tuning procedure
  - Inner folds: tune hyperparameters on each outer-fold training set via cross-validation
- OOF evaluation: inspect out-of-fold predictions for instability and bias
- **Conclusion:** tuning generalises stably $\rightarrow$ proceed to full-data tuning + refit

Final models are refit on *all* data with optimal hyperparameters — no held-out test set, because the goal is understanding, not out-of-fold performance.

OOF Diagnostics — XGBoost (MI + VIF, 4 features)



OOF Diagnostics — XGBoost (VIF only, 6 features)



GAM Variant Selection

Three candidates:

- **LinearGAM** — fitted on raw `ml_ade`
- **LinearGAM (log)** — fitted on $\log(1 + \texttt{ml\_ade})$, evaluated on original scale
- **GammaGAM** — Gamma distribution with log link, suited for positive right-skewed targets

Selection criterion: lowest mean outer-fold RMSE (original scale).

Variant	RMSE	R^2	MAE
LinearGAM	**0.440**	**0.515**	0.268
LinearGAM (log)	0.444	0.506	**0.255**
GammaGAM	0.475	0.434	0.277

Outer-fold means, 3-fold nested CV.
LinearGAM selected.

Final Model Performance (OOF)

GAM (LinearGAM)

- Fitted on raw `ml_ade`
- Additive smooth terms, one per feature

Metric	OOF value
RMSE	0.440
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XGBoost

- Gradient-boosted trees
- Fitted on `log1p(ml_ade)`
- Post-hoc interpretation via SHAP

Metric	OOF value
RMSE	**0.387**
R^2	**0.624**
MAE	**0.213**

XGBoost outperforms GAM on all metrics. Both models are used as a cross-check: the feature-effect narratives should agree across approaches.

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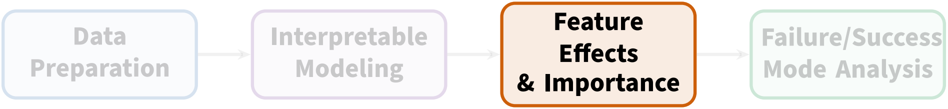

Global Feature Importance

XGBoost — mean $|\mathrm{SHAP}|$

Feature	**Mean** $|\mathrm{SHAP}|$
`std_speed`	**0.129**
`heading_change_per_sec`	0.070
`min_neighbor_distance`	0.027
`mean_acceleration`	0.027
`scene_density_VEHICLE`	0.020
`scene_num_VEHICLE`	0.016

GAM — smooth-term significance

Feature	$p < 0.05$**?**
`heading_change_per_sec`	✓
`mean_acceleration`	✓
`min_neighbor_distance`	✓
`std_speed`	✓
`scene_num_VEHICLE`	✓
`scene_density_VEHICLE`	×

Both models agree: speed variation (`std_speed`) and turning (`heading_change_per_sec`) are the dominant drivers of prediction error. Scene density is consistently least important.

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Motion features

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Scene & proximity features

- `min_neighbor_distance`: mild positive effect — isolated agents are marginally harder to predict
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- `scene_density_VEHICLE`: not significant in GAM; weakest SHAP contribution in XGBoost

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$\Rightarrow$ Trajectron++ struggles most with **fast, erratic, turning pedestrians**; scene context is a secondary factor.

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- Global picture consistent with GAM: `std_speed` contributes most strongly
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- Confidence bands narrow in high-density regions

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Data Preparation → Interpretable Modeling → Feature Effects & Importance → **Failure/Success Mode Analysis**

Performance Regime Methodology

1. Divide trajectories into difficulty groups based on `ml_ade` (easy / medium / hard)
2. Run clustering on trajectory + scene characteristics per group
   - Option A: original feature space
   - Option B: UMAP-reduced space (dimensions chosen via trustworthiness score + elbow rule)
3. Assess clustering quality with DBCV score
4. Export identified clusters (performance regimes)

Clustering Results

Performance group sizes (VIF-only run)

Group	Boundary	N	%
Easy	< 0.39	15 348	59%
Medium	0.39–1.20	8 094	31%
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Cluster quality by group

Group	**DBCV**	**Noise %**
Easy	**0.411**	30%
Medium	0.397	40%
Hard	0.091	75%

Easy group has the clearest substructure.
Hard is mostly a continuous high-error regime.

Regime Inspection — What Drives Error in Each Regime?

Easy regime

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Hard regime

- 99.5% of hard trajectories have *positive* total feature effect
- Dominated by high `std_speed` ($+2.0\sigma$ global), turning, and acceleration

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- 99.5% of hard trajectories have *positive* total feature effect
- Dominated by high `std_speed` ($+2.0\sigma$ global), turning, and acceleration
- No sharp sub-clusters — hard is a **continuous high-variation regime**; a small extreme outlier group has `std_speed` $\approx +4.8\sigma$

Takeaway: Trajectron++ struggles with erratic, fast-turning pedestrians. Slow or near-stationary pedestrians are easy regardless of scene complexity.

Outline

Settings Sweep Setup

- Dataset: nuScenes mini (full dataset too expensive for repeated runs)
- Cartesian sweep over:
  - `history_sec`
  - `prediction_sec`
  - `attention_radius_m`
- Pooled model: model setting variables + trajectory/scene features as control variables
- Feature normalization: setting-agnostic versions to avoid mechanical confounding (e.g. `mean_path_length_per_second` instead of `path_length`)

Effect of Model Settings

XGBoost — mean $|\text{SHAP}|$ (pooled model)

Feature	Mean $|\text{SHAP}|$
`prediction_sec`	**0.178**
`attention_radius_m`	0.098
`std_speed`	0.093
`max_speed`	0.070
`history_sec`	0.022

All 8 features significant in GAM ($p \ll 0.01$).

Key findings

- `prediction_sec`: largest effect — longer horizon means harder forecasting (ADE grows with time)

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- Trajectory features (`std_speed`, `max_speed`) retain strong explanatory power even after controlling for settings

Outline

Summary of Findings

1. **RQ1 — Which characteristics drive prediction error?**
   - `std_speed` (speed variation) and `heading_change_per_sec` (turning) are the dominant drivers in both GAM and XGBoost
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3. **RQ3 — How do model settings affect performance?**
   - Prediction horizon (`prediction_sec`) has the largest effect — longer horizon raises error strongly
   - Attention radius (`attention_radius_m`) is second; history window (`history_sec`) has minimal impact

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- Hard-regime clustering is weak (75% noise); discrete failure-mode identification remains an open challenge
- Settings sweep uses nuScenes *mini* — results may not transfer to full-scale training

Future Work

- Extend pipeline to vehicle and cyclist trajectories
- Apply to other trajectory prediction models and datasets via the trajdata interface
- Explore higher-capacity meta-models to better isolate failure modes in the hard regime